Algebraic combinatorics

Lecture 17: The Fundamental Lemma of (P, w)-partitions (20/11/2025)

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WARNING: any information contained in these notes can be wrong!
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Contents

1. Labelled posets and (P, w)-partitions

We're looking to find a suitable statistic to associate to $\Omega(P, n)$, the order-reversing[1] functions from a poset P to a chain of length n (e.g., $\{1 \le 2 \le \cdots \le n\}$).

As a first approach, we will define a generating function tracking *every* aspect of such functions, and we will restrict ourselves to a more particular statistic in the future lectures.

In order to do so, we will assume we have a fixed *labelling* w for our poset P:

Definition 1. Let P be a poset with n elements. A *labelling of P* is a bijection $w : P \to [n]$.

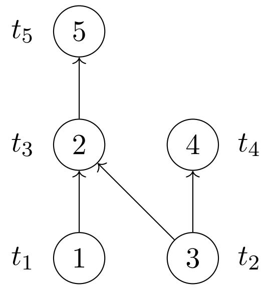

Figure 1. A *labelled* poset $\{t_1, t_2, t_3, t_4, t_5\}$, with t_1, $t_2 \le t_3 \le t_5$ and $t_2 \le t_4$. The labels are represented within the circled nodes.

[1]This is merely a convention. The theory concerning order-preserving maps is studied using essentially the same arguments, with appropriate adjustments.

We are now ready to state the definition for a (P,w)-partition, which allows us to be more flexible on the choice of the "weaknesses" of our order-reversing maps:

Definition 2. A (P,w)-partition is a map $\sigma : P \to \mathbb{N}$ with the following two properties:

(i.) *it's (weakly) order-reversing:* if $s <_P t$, then $\sigma(s) \geq \sigma(t)$;

(ii.) *it respects the strictness according to the labelling:* if $s <_P t$ and $w(s) > w(t)$, then $\sigma(s) > \sigma(t)$.

Notation. We will denote the set of all (P,w)-partitions of P with $A(P,w)$.

Example 1.1. For Figure 1, the resulting conditions for a map σ to be a (P,w)-partition are:

- being (weakly) order-reversing;
- $\sigma(t_2) > \sigma(t_3)$, since $t_2 < t_3$ and $w(t_2) = 3 > 2 = w(t_3)$.

Remark 1.1. If we choose a labelling w such that, whenever $s <_P t$, $w(s) < w(t)$, a (P,w)-partition is simply an order-reversing map – it must satisfy only (i.), since (ii.) is vacuously true.

Likewise, if we choose w such that $w(s) > w(t)$, then the map must be *strictly* order-reversing.

These observations lead us to the following definitions:

Definition 3. We say a labelling w is *natural* if, whenever $s <_P t$, $w(s) < w(t)$. We say w is *dually natural* if $w(s) > w(t)$ whenever $s <_P t$.

For (dual) natural labellings, the conditions for a map to be (P,w)-partition do not depend on w. Therefore they are simply called (strict) P-partitions.

We are now ready to define a statistics on the (P,w)-partitions:

Definition 4. Let $P = \{t_1, t_2, \dots, t_n\}$ be a poset with labelling w. We then define the (P,w)*-partition enumerator* as:

$$F_{P,w} := F_{P,w}(x_1, \dots, x_n) := \sum_{\sigma \in A(P,w)} x_1^{\sigma(t_1)} \cdots x_n^{\sigma(t_n)}.$$

Remark 1.2. Given a natural labelling w, the generating function $F_{P,w}$ enumerates all the order-reversing maps from the $\Omega(P,n)$'s. To restrict to a certain n, we simply ignore any monomials that contain a variable with degree greater than n.

The following examples provide insight into why the (P,w)-partition enumerator, in a certain sense, generalizes the partitions we've studied on the integers.

Example 1.2. Let P be a naturally labelled chain $t_1 < t_2 < \cdots < t_n$. Recall that in this case a P-partition is the same as an order-reversing map. Thus:

$$F_P = \sum_{a_1 \geq a_2 \geq \cdots a_n \geq 0} x_1^{a_1} \cdots x_n^{a_n} = \frac{1}{(1-x_1)(1-x_1x_2)\cdots(1-x_1x_2\cdots x_n)}.$$

Example 1.3. Let P be a dually naturally labelled chain $t_1 < t_2 < \cdots < t_n$. In this case a P-partition is the same as a *strict* order-reversing map. Therefore:

$$F_P = \sum_{a_1 > a_2 > \cdots a_n > 0} x_1^{a_1} \cdots x_n^{a_n} = \frac{x_1^{n-1} x_2^{n-2} \cdots x_{n-1}}{(1-x_1)(1-x_1x_2)\cdots(1-x_1x_2\cdots x_n)}.$$

Example 1.4. Let P be an antichain (i.e., no comparisons are possible) of size n. It is a simple **exercise** to check that (P, w)-partitions do not depend on w and are simply maps from P to $\mathbb{N}$. Hence:

$$F_P = \frac{1}{(1 - x_1)\cdots(1 - x_n)}.$$

Remark 1.3. F_P enumerates, in a certain sense, the possible partitions on $[n]$. Choosing $x_i = x^i$, we get the *partition enumerator* series.

Notation. Let P a poset of size n with labelling w. We will employ the notation $\mathcal{L}(P, w)$ to denote the set of permutations $\pi = \pi_1 \cdots \pi_n \in S_n$ such that the map $\sigma : P \to [n]$ defined by $\sigma(w^{-1}(\pi_i)) = i$ is a linear extension (i.e., an order-preserving map) of P.

Remark 1.4. Let $P = [n]$ be a poset. There is a natural labelling on P, namely the identity map id $\in S_n$. This simplifies the definition of $\mathcal{L}(P, w)$:

> $\mathcal{L}(P, w)$ is the set of permutations $\pi = \pi_1 \cdots \pi_n \in S_n$ such that the map $\sigma : P \to [n]$ defined by $\sigma(\pi_i) = i$ is a linear extension (i.e., an order-preserving map) of P.

In other words:

> $\mathcal{L}(P, w)$ is the set of permutation $\pi \in S_n$ such that if $i \leq_P j$, then i comes before j in the one-line notation of π.

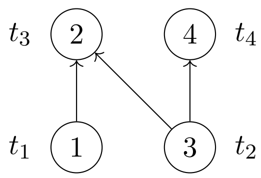

FIGURE 2. A *labelled* poset $\{t_1, t_2, t_3, t_4\}$, with t_1, $t_2 \leq t_3$ and $t_2 \leq t_4$. The labels are represented within the circled nodes.

Example 1.5. Let's compute $\mathcal{L}(P, w)$ for the labelled poset from Figure 2. 1 and 3 must come before 2; 3 must come before 4 as well. Therefore:

$$\mathcal{L}(P, w) = \{1324, 1342, 3124, 3142, 3412\}.$$

2. π-COMPATIBILITY AND THE FUNDAMENTAL LEMMA OF (P, w)-PARTITIONS

Definition 5. Let π be a permutation of n elements. A function $f : [n] \to \mathbb{N}$ is said to be *π-compatible* if:

- *it weakly decreases along the one-line notation of* π*:* $f(\pi_1) \geq f(\pi_2) \geq \cdots \geq f(\pi_n)$.
- *strictness is forced upon a descent:* $f(\pi_i) > f(\pi_{i+1})$ if i is a descent (i.e., if $\pi_i > \pi_{i+1}$).

Lemma 2.1. *Let f be a function from $[n]$ to $\mathbb{N}$. Then, there exists a unique permutation $\pi \in S_n$ such that f is π-compatible.*

Proof. Consider the unique (**exercise**) ordered set partition $(B_1, B_2, \ldots, B_k)$ such that $f\big|_{B_i}$ is constant and $f\big|_{B_1} > f\big|_{B_2} > \cdots > f\big|_{B_k}$. (continue)

Example 2.1. Let $f : [5] \to \mathbb{N}$ be such that:

- $f(2) = f(3) = 1$;
- $f(1) = f(4) = 6$;
- $f(5) = 8$.

Thus, $B_1 = \{5\}$, $B_2 = \{1, 4\}$ and $B_3 = \{2, 3\}$, since $8 > 6 > 1$.

We are now ready to construct our permutation π by specifying its one-line notation. First, we order the elements within each block B_i in increasing order:

$$B_i = \{b_{i,1} \leq b_{i,2} \leq \cdots \leq b_{i,j_i}\}.$$

We then define π by concatenating these ordered blocks:

$$\pi = b_{1,1} \cdots b_{1,j_1} b_{2,1} \cdots b_{2,j_2} \cdots b_{k,j_k}.$$

One can verify that f is indeed π-compatible (**exercise**). The uniqueness of the permutation π follows straightforwardly from the uniqueness of the set partition $(B_1, B_2, \ldots, B_k)$. □

Definition 6. Let σ be a map from P to $\mathbb{N}$, where (P, w) is a labelled poset of size n. We then define the map σ' from $[n]$ to $\mathbb{N}$ to be the re-indexed version of σ via w, where the label $w(p) \in [n]$ is used instead of the poset element $p \in P$. In other words, σ' is such that:

$$\sigma'(i) = \sigma(w^{-1}(i)).$$

Notation. Let (P, w) be a labelled poset of size n, and let $\pi \in S_n$ be a permutation of n elements. We will employ the notation S_π to denote the set of all maps σ from P to $\mathbb{N}$ such that their re-indexed versions σ' are π-compatible.

Theorem 2.2 (Fundamental Lemma of (P, w)-partitions)**.** *The set of (P, w)-partitions $A(P, w)$ is exactly the disjoint union of the S_π's, where π varies over $\mathcal{L}(P, w)$:*

$$A(P, w) = \bigsqcup_{\pi \in \mathcal{L}(P, w)} S_\pi.$$

Proof. The disjointness of the right hand side follows straightforwardly from Lemma 2.1. We now prove both inclusions.

($\subseteq$) Let $\sigma \in A(P, w)$. Lemma 2.1 tells us there exists a unique π such that $\sigma \in S_\pi$. It remains only to verify that π is an element of $\mathcal{L}(P, w)$.

Let $\pi = \pi_1 \cdots \pi_n$. Suppose $i < j$, $w(s) = \pi_i$ and $w(t) = \pi_j$. We need to show that $s \not> t$.

Since $\sigma \in S_\pi$, then:

$$\sigma(s) = \sigma'(\pi_i) \geq \sigma'(\pi_j) = \sigma(t).$$

If we had $\sigma(s) > \sigma(t)$, then $s \not> t$, since σ is order-reversing. Suppose then $\sigma(s) = \sigma(t)$, i.e., $\sigma'(\pi_i) = \sigma'(\pi_j)$. Therefore we have:

$$\pi_i < \pi_{i+1} < \cdots < \pi_j,$$

since there cannot be descents (otherwise we'd have $\sigma(s) > \sigma(t)$, **exercise**). Thus $w(s) = \pi_i < \pi_j = w(t)$: if we had $s > t$, we would have $\sigma(t) > \sigma(s)$, which contradicts our supposition. It follows that s cannot be greater than t, and then that $A(P,w) \subseteq \bigsqcup_{\pi\in\mathcal{L}(P,w)} S_\pi$.

($\supseteq$) Let σ be a map from P to $\mathbb{N}$ such that σ' is π-compatible, with $\pi \in \mathcal{L}(P,w)$. We need to show that σ is an element of $A(P,w)$. We will do so by showing that σ satisfies the two conditions for being a (P,w)-partition.

Let s, t be elements of P such that $s < t$. Let $\pi_i = w(s)$ and $\pi_j = w(t)$. Since π is an element of $\mathcal{L}(P,w)$, then π_i must come before π_j in the one-line notation of π, i.e., $i < j$. Since σ' is π-compatible, we have $\sigma(s) = \sigma'(\pi_i) \geq \sigma'(\pi_j) = \sigma(t)$.

Suppose now that $w(s) > w(t)$. Then, π_i is strictly bigger than π_j, meaning there exists a descent k with $i \leq k < j$. Therefore:

$$\sigma(s) = \sigma'(\pi_i) \geq \sigma'(\pi_{i+1}) \geq \cdots \geq \sigma'(\pi_k) > \sigma'(\pi_{k+1}) \geq \cdots \geq \sigma'(\pi_j) = \sigma(t).$$

This concludes our proof.

□

The usefulness of the Fundamental Lemma of (P,w)-partitions comes from the fact that the generating function $F_{P,w}$ is easier to describe on S_π, as shown in the following result.

Definition 7. Let $\pi \in S_n$ be a permutation of n elements, and let $P = \{t_1, \ldots, t_n\}$ be a poset with size n and labelling w. We then define a new permutation $\pi' \in S_n$, such that π' gives the index of the element in P corresponding to the value in π. Specifically, π' is such that:

$$\pi_i = w(t_j) \implies \pi'_i = j.$$

Lemma 2.3. *Let $\pi \in S_n$ be a permutation of n elements, and let $P = \{t_1, t_2, \ldots, t_n\}$ be a poset with size n and labelling w. Then:*

$$F_{P,w,\pi} := \sum_{\sigma\in S_\pi} x_1^{\sigma(t_1)} \cdots x_n^{\sigma(t_n)} = \frac{\prod_{j\in\mathrm{Des}(\pi)} x_{\pi'_1} x_{\pi'_2} \cdots x_{\pi'_j}}{\prod_{i=1}^{n}(1 - x_{\pi'_1} \cdots x_{\pi'_i})}.$$

Proof. <u>This part is happily left to the professor to fill.</u> □

As an immediate application of Lemma 2.3 and the Fundamental Lemma of (P,w)-partitions, we get the following theorem (**exercise**):

Theorem 2.4. *Let $P = \{t_1, t_2, \ldots, t_n\}$ be a poset with size n and labelling w. Then:*

$$F_{P,w} = \sum_{\pi\in\mathcal{L}(P,w)} \frac{\prod_{j\in\mathrm{Des}(\pi)} x_{\pi'_1} x_{\pi'_2} \cdots x_{\pi'_j}}{\prod_{i=1}^{n}(1 - x_{\pi'_1} \cdots x_{\pi'_i})}.$$

3. Applications of the Fundamental Lemma

Let's specialise the series from Theorem 2.4 and get its q-analogue. We will denote it with $F_{P,w}(q) := F_{P,w}(q,q,\ldots,q)$.

Example 3.1. Let P be a chain of size n with a natural labelling. Thus $\mathcal{L}(P,w)$ consists of only the identity $\mathrm{id} \in S_n$, with $\mathrm{maj}(\mathrm{id}) = 0$. Therefore:

$$F_P(q) = \frac{q^{\mathrm{maj}(\pi)}}{(1-q)(1-q^2)\cdots(1-q^n)} = \frac{1}{(1-q)(1-q^2)\cdots(1-q^n)},$$

which is consistent with what we'd already seen in Example 1.2

The Fundamental Lemma of (P,w)-partitions and Theorem 2.4 allow us to provide alternative proofs for some of the theorems established in previous lectures.

Example 3.2. Let P be an antichain of size n. Thus $\mathcal{L}(P,w) = S_n$. Using the results from Example 1.4, we get:

$$\frac{1}{(1-q)^n} = F_P(q) = \frac{\sum_{\pi\in S_n} q^{\mathrm{maj}(\pi)}}{(1-q)(1-q^2)\cdots(1-q^n)},$$

from which the MacMahon theorem follows:

$$\sum_{\pi\in S_n} q^{\mathrm{maj}(\pi)} = \prod_{i=1}^{n} \frac{1-q^i}{1-q} = \prod_{i=1}^{n} [i]_q = [n]_q!\,.$$

Example 3.3. Let $P = C_n + C_k$ be a disjoint union of two chains (see Figure 3).

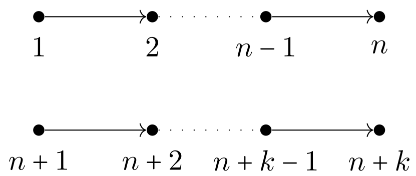

Figure 3. The *labelled* disjoint union of two chains, one of length n and the other of length k.

One can easily verify (**exercise**) that:

$$F_P(q) = F_{C_k}(q) \cdot F_{C_n}(q).$$

Recall from Example 3.1 that the following identity holds:

$$F_{C_n}(q) = \prod_{i=1}^{n} \frac{1}{1-q^i}, \qquad F_{C_k}(q) = \prod_{j=1}^{k} \frac{1}{1-q^j}.$$

We can identify $\mathcal{L}(P)$ with words with n 0's and k 1's, with the zeros representing elements from C_n and the ones representing elements from C_k (**exercise**). Thus, by applying Theorem 2.4 we get:

$$\left(\prod_{i=1}^{n} \frac{1}{1-q^i}\right)\left(\prod_{j=1}^{k} \frac{1}{1-q^j}\right) = \frac{\sum_{\omega\in R(0^n1^k)} q^{\mathrm{maj}(\omega)}}{\prod_1^{n+k}(1-q^i)},$$

from which the following identity follows again (**exercise**):

$$\begin{bmatrix} n+k \\ k \end{bmatrix}_q = \sum_{\omega \in R(0^n 1^k)} q^{\mathrm{maj}(\omega)}.$$